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Environmental impacts of food waste in Europe

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ABSTRACT

Approximately 88 Million tonnes (Mt) of food is wasted in the European Union each year and the environmental impacts of these losses throughout the food supply chain are widely recognised. This study illustrates the impacts of food waste in relation to the total food utilised, including the impact from food waste management based on available data at the European level. The impacts are calculated for the Global Warming Potential, the Acidification Potential and the Eutrophication Potential using a bottom-up approach using more than 134 existing LCA studies on nine representative products (apple, tomato, potato, bread, milk, beef, pork, chicken, white fish). Results show that 186 Mt CO_2-eq, 1.7 Mt SO_2-eq. and 0.7 Mt PO_4-eq can be attributed to food waste in Europe. This is 15 to 16% of the total impact of the entire food supply chain. In general, the study confirmed that most of the environmental impacts are derived from the primary production step of the chain. That is why animal-containing food shows most of the food waste related impacts when it is extrapolated to total food waste even if cereals are higher in mass. Nearly three quarters of all food waste-related impacts for Global Warming originate from greenhouse gas emissions during the production step. Emissions by food processing activities contribute 6%, retail and distribution 7%, food consumption, 8% and food disposal, 6% to food waste related impacts. Even though the results are subject to certain data and scenario uncertainties, the study serves as a baseline assessment, based on current food waste data, and can be expanded as more knowledge on the type and amount of food waste becomes available. Nevertheless, the importance of food waste prevention is underlined by the results of this study, as most of the impacts originate from the production step. Through food waste prevention, those impacts can be avoided as less food needs to be produced.

1. Introduction

Food production is one of the key contributors to consumption-related environmental impacts in the European Union (EU). The primary production of food requires the use of resources such as fuels, land, water and raw materials that have associated economic and environmental impacts. Studies have shown that food contributes about 20 to 30% of the total impact of private consumption (Tukker et al., 2006). A major source of impacts is agricultural processes such as fertilizer application or livestock farming. Livestock farming produces significant environmental emissions in the form of methane derived from the enteric fermentation of ruminants. The application of fertilisers creates direct emission of nitrous oxides from soil processes. Both emissions contribute to climate change as do emissions related to energy for transport, storage and cooking of food in other steps of the food supply chain. Moreover, due to fertilizer application, ammonia is released as well as nitrogen oxides after the denitrification process in soils, which contributes to acidification and eutrophication. Emissions from manure storage and combustion processes also add to these environmental impacts.

The environmental impact of food production and consumption is further exacerbated when food is wasted rather than consumed. The FAO estimates that globally one third of food produced for human consumption is lost or wasted throughout the entire supply chain (FAO, 2011). Current estimates for the EU show that 88 Million tonnes (Mt) (±14 Mt) of food waste is produced along the supply chain, which is equivalent to 173 kg ± 27 kg per capita and year (Stenmarck et al., 2016). The environmental impacts of food waste cover all emissions deriving from the different steps of the food supply chain. The later in the supply chain a product is wasted, the higher are its environmental impacts, since all emissions coming from up-stream steps of the supply chain (e.g. production,

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processing, transport etc.) are included in the overall impact of the waste material. In other words, when food is wasted, then all associated activities and emissions created in the food supply chain upstream are therefore in vain. The environmental impact of food waste throughout the supply chain and subsequent waste disposal is considerable. Avoidable waste from the food supply chain results in unnecessary environmental impact from overproduction and processing upstream. Reducing avoidable wastes could reduce the amount of food production and it associated impacts overall. The UN Sustainable Development Goal 12.3 which targets a 50% reduction in food waste at the retail and consumer levels, in addition to reducing food losses along production and supply chains by 2030 can therefore serve as a significant step towards a reduction of environmental impacts due to food waste.

To understand the overall impacts of food waste and therefore to detect hotspots for future prevention activities, the impacts of waste throughout the supply chain as well as for different food products need to be determined more in detail. Studies on the environmental impacts of food consumption baskets (e.g. in Eberle and Fels, 2015; Foster et al., 2006; Notarnicola et al., 2017) have already provided some indications of hotspots in relation to food products. Those studies also highlighted the contribution of food waste to overall environmental impacts since food waste can be a high percentage of the initial weight depending on the specific food products. Consumption related studies have concluded that meat and dairy products make their major contribution to the environmental impacts in the agricultural sector (Audsley et al., 2009; Foster et al., 2006; Leip et al., 2015; Tukker and Jansen, 2006). The impacts from livestock farming contribute about 10% of the total GHG emissions from the EU-27 (Lesschen et al., 2011). In contrast, supply chain studies have shown that food consumption is of great importance regarding environmental emissions from vegetal food products (Foster et al., 2006). Indeed, the influence of the consumer behaviour (e.g. the way of organising shopping trips, storing of food in the refrigerator, energy demand for cooking) on the LCA results was identified as being highly relevant (Gruber et al., 2015).

Relatively few studies include waste management operations for food waste for the determination of environmental impacts. Food waste management can differ from country to country, and also from step to step of the supply chain, depending on the waste management strategy in use and also the type of food (vegetal or animal-derived). Different food waste management operations and their environmental impacts were assessed in Gruber et al. (2015) and Eriksson et al. (2015).

A range of studies have already addressed the issue of food waste and its environmental impacts in a national context (e.g. for UK in Quested and Johnson, 2009), for specific products (e.g. for tomato in Bernstad et al., 2017) or for different steps of the chain (Scholz, 2013; Brancoli et al., 2017; Quested and Johnson, 2009). Two studies are even available on a European (Monier et al., 2010) and a global level (FAO, 2013). Those existing studies especially on the bigger European or global level very often lack details on specific issues identified as important (e.g. waste management, consumer behaviour or even detailed information on specific food products). Therefore, within this study, a comprehensive approach was chosen; we have combined all relevant issues by using the most recent European food waste data in combination with an in-depth literature review to generate environmental impacts on food category level including waste management.

The goal of this study was to estimate the impact of food waste in relation to the total food utilised[1] in a European context, including the impact of food waste management based on current food waste data sets. The assessment of environmental impacts was performed under the EU project FUSIONS (Food Use for Social Innovation by Optimising Waste Prevention Strategies) which sought to contribute towards achieving a more resource efficient Europe through the significant reduction in food waste. Two approaches were used in FUSIONS, the bottom-up and top-down, to determine the overall impacts of food and food waste with a preliminary food waste data set elaborated in the same project (see Scherhaufer et al., 2015). This study comprises the findings of FUSIONS and conducts the assessment from the bottom-up on the basis of the food waste data from Stenmarck et al. (2016).

2. Methodology

2.1. Definition of food waste

The definition of food waste is often controversial and many different definitions have been developed (Schneider, 2013). To aid transparency, the system boundaries of the study need to be clearly described. Food waste can be classified into various categories (edible/inedible, avoidable/unavoidable). The classification into avoidable and unavoidable food waste is quite common, but still it is used inconsistently (Lebersorger and Schneider, 2011). A distinction between avoidable and unavoidable food waste is necessary to estimate the reduction potential of food waste by waste prevention (Huber-Humer et al., 2017). A harmonization of the definition of food waste is required and also set as a goal in the EU Action Plan for the Circular Economy (European Commission, 2015). In this study the term 'food waste' is that defined by Östergren et al. (2014). Food waste in this context covers both edible and inedible parts of food removed from the food supply chain excluding food used for biochemical and bio-based processes or for use as animal feed (Östergren et al., 2014).

2.2. System boundary

The assessment in the current study covers all impacts from cradle to grave, meaning all steps of the food life cycle including primary production covering agriculture, aquaculture and fisheries (PP), food processing and manufacturing (FP), retail and distribution including transport processes and packaging (RD) as well as food preparation and consumption (including out-of-home consumption) (FC) and food disposal (FD) at each step of the chain. Food waste is defined as food and inedible parts removed from the food supply chain which are not used for food conversion or valorisation (e.g. animal feed) (see Östergren et al., 2014). Food removed from the supply chain and fed to animals or used for biobased materials or biochemical processing, are considered as by-products or co-products in the current study. The end of life step of the food life cycle is defined in this study as food disposal (FD). Food disposal covers all waste management operations such as composting and anaerobic digestion, co-generation, incineration, landfilling but also other routes such as production of bio-energy and disposal via the sewer, plough-in/not harvested and any discards.

The food life cycle is furthermore classified into impacts deriving from different steps of the supply chain and impacts deriving from food which is eaten and food which is wasted. Fig. 1 shows the structure of the food life cycle and terms used in this study on the example of an apple. As the purpose of the product being assessed in this study is to eat it and not to waste it, the functional unit was chosen to be the amount of food eaten by an EU citizen. The mass flows are referred to this unit.

It is explained on the example of 1 kg of food (in this case apple) eaten by a consumer and illustrated in Fig. 1: Food waste related emissions cover all emissions which are released in

[1] Defintion by FAOStat: Production + imports – exports + changes in stocks (decrease or increase) = supply for domestic utilization

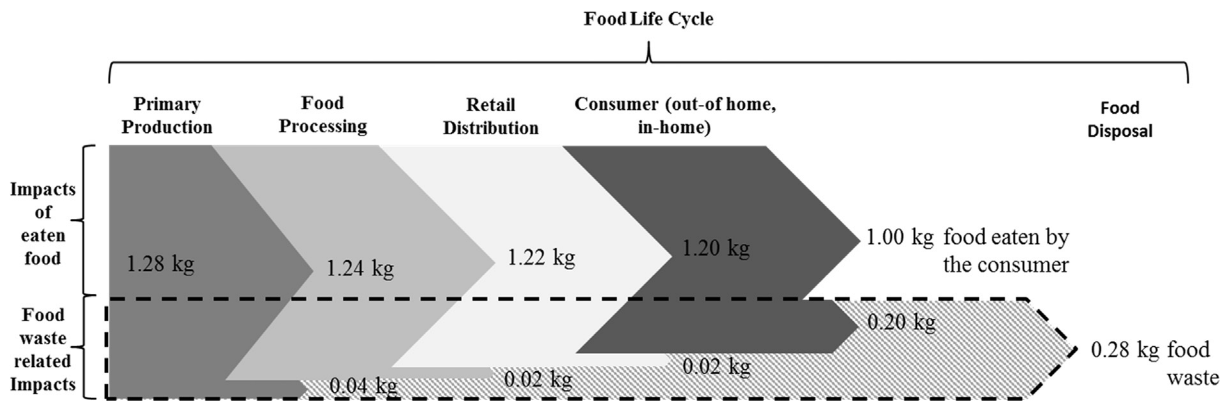

Fig. 1. System boundaries of the environmental assessment and functional unit (based on the example of an apple).

production (grey arrow), processing (light grey arrow), retail and distribution (white arrow) and consumption (dark grey arrow) of this wasted food including all emissions due to waste operations and disposal (patterned arrow). The supply chain starts with the primary production of 1.28 kg of apples, where 0.04 kg are removed from the step of the chain as food waste, then the remaining 1.24 kg of food are processed, where another 0.02 kg are removed as food waste again, then 1.22 kg of food are distributed and so on. Finally, 1 kg of apples is finally eaten by consumers and a total of 0.28 kg of apples is wasted along the whole supply chain. In other words, for every 1 kg of apples eaten by the consumer, 1.28 kg would need to be produced. Beside the environmental impact at different stages in the supply chain, the emissions for the disposal of the 0.28 kg wasted apples are accounted to the food waste related emissions (patterned arrow). Composting along with further use of compost, anaerobic digestion with energy recovery, waste incineration with and without energy recovery, landfilling with gas collection but without gas utilization as well as discarded waste to land were considered as options for food disposal. Human excretion and waste water treatments deriving from food which is eaten are beyond the system boundaries of this study.

This study follows an attributional approach with system expansion to solve multi-functionality and includes the waste disposal operations, which is in line with the ILCD guidelines (see p. 43 in European Commission, 2010). For crediting material recovery of compost and digestate, virgin production processes were considered to be substituted (in this case conventional fertilizer and peat). For crediting energy recovery, the substitution of average EU-27 electricity mix and average EU-27 district heating mix was considered.

2.3. The bottom-up approach

Both bottom-up and top-down approaches can be found in the literature in relation to food LCAs. In the bottom-up approach, existing product LCAs are extrapolated whereas in the top-down approach environmentally extended input-output tables are used and allocated to a certain unit (Tukker et al., 2006). Both approaches have their own advantages and disadvantages in use. The bottom-up approach is commonly used in studies to detect priority product categories for consumption (see Jansen and Thollier, 2006). In this study it is used to detect priority product categories in food waste. An extrapolation to total food waste was furthermore possible with this approach. However, both approaches were conducted in parallel within the EU project FUSIONS (Scherhaufer et al., 2015), which makes a comparison of the results possible (see in Section 5).

The bottom-up approach uses existing product LCAs for a selected number of products as being representative of a product category (in this study called indicator products). The current study is therefore not a comparative LCA according to ISO Standards (ISO, 2006a, 2006b), but uses LCA results for different products. The bottom-up approach is literature based which makes a comprehensive literature review necessary to cover different production schemes or waste management options.

The current study covered a detailed screen of literature data to enable a data collection on a life cycle step level. In total 134 LCA studies, which cover one or even more of the selected indicator products, were collected in a database. A full list of studies used in the database is shown in Scherhaufer et al. (2015). Although a considerable amount of studies was available, data gaps were identified which need to be filled. The consumption phase (incl. consumer travel) is very heterogeneous in its nature and also in existing LCA studies, which makes a harmonization of data necessary. For this purpose, in-house calculations on e.g. certain cooking and storage behaviours were carried out to be consistent over the whole data inventory. A major data gap was found concerning waste management operations. For example, manure management was only included in some studies. To fill this gap, a determination of the share of different waste management operations used for food waste in all life cycle stages was necessary. Data inventories taken from LCA studies and also in-house calculated datasets, both on consumer as well as waste management level, are shown in detail in Section 3.

2.4. Indicator products

The variety of food products is enormous and a complete picture of its environmental impacts for all consumed articles is not currently possible. Previous studies have shown that there are notable differences in terms of environmental impacts between e.g. meat or dairy products and vegetables or fruits. Therefore, the bottom-up approach applied to specific indicator products was chosen as a method to estimate the environmental impacts of food and food waste in Europe. Within this approach each indicator product represents certain food categories and these food categories further represent the whole food supply chain. Indicator products were chosen on a mass basis from consumption data which provide EU-averages such as FAOSTAT (FAOSTAT, 2011), EUROSTAT (Eurostat, 2011) and EFSA (EFSA). Also, country specific datasets were conducted as decision support such as two Finnish datasets "Findiet" and "Balance sheet for food commodities" (Helldán et al., 2013; Tike, 2014), British dataset on food purchases (Defra, 2013), Swedish "Food consumption data" (Jordbruksverket, 2013), Irish "National adult nutrition survey" (Walton, 2011) and

Dutch "National food consumption survey" (van Rossum et al., 2011). The survey resulted in the following indicator products being chosen for the environmental assessment: apple, tomato, potato, bread, milk, beef, chicken, pork, and white fish, where, apple represents 'apples and products'-category (e.g. not only fresh apple but also apple products such as apple marmalade) and so on (see Table 1).

To show the representativeness of the chosen products, production data from FAOStat, which provides EU-averages and country specific information on production amounts, imports and exports of food commodities, were considered. Regarding food consumption data, it was also decided to use FAOStat data on food domestic utilization. This was because FAOStat data was considered to be the most consistent data, avoiding data gaps and double counting. Additionally, consumption datasets were considered as inadequate since most of the data found was over 10 years old with relatively few recent datasets (from year 2007 onwards). Table 2 shows the total amount of food production and food domestic utilization in EU 2011 (excl. alcohol) per food category and the share of indicator products within the respective food category. It shows that most of the food categories can be represented by the chosen indicator products to a reasonable extent. Only fruits and vegetables are covered with a share of lower than 30%. This is because of the wide variety of fruits and vegetables within this category. It is however assumed that the products chosen are similar to products within the food category when it comes to environmental impacts (e.g. the impact of a pear may be comparable to apples). Overall the indicator products represent 63% of food production and 65% of food domestic utilization within the EU. The missing food categories include oil crops, pulses, sugar crops and nuts as well as some small categories such as miscellaneous. As food waste is also produced in those product categories it needs to be considered in further studies to detect their environmental impacts.

There is a wealth of research papers comparing the environmental impact of organic vs. non-organic food production. However, given the relatively low level of sales of organic food in European countries (on average 3.9% of sales for 2014 according to IFOAM EU et al., 2016), only non-organic products were chosen as indicator products.

Table 1
Indicator products and their representing product category.

No.	Indicator product	Indicator product category
1	Apple	Apples and products
2	Tomato	Tomatoes and products
3	Potato	Potatoes and products
4	Bread	Cereals
5	Milk	Milk products
6	Beef	Bovine meat
7	Chicken	Poultry meat
8	Pork	Pig meat
9	White fish	White fish (demersal)

2.5. Calculation method

For the calculation of environmental impacts of each indicator product in each step of the supply chain and to generate food waste related emissions along the supply chain, data on the following parameters were determined:

- Mass flow of eaten food (m_F) for each of the nine indicator products (1–9): $m_{F1\text{-}9}$ ... the amount of food eaten by the consumer in mass for each of the nine indicator products
- Mass flow of food waste (m_{FW}) in each step of the chain ($m_{FW/PP}$, $m_{FW/FP}$, $m_{FW/RD}$, $m_{FW/FC}$, $m_{FW/FD}$) and each of the nine indicator products (1–9): e.g. $m_{FW1\text{-}9/PP}$ ...... the amount of food waste in mass which is generated in the primary production step of the chain for each of the nine indicator products
- Environmental impacts (I) from existing LCA studies for each step of the chain (I_{PP}, I_{FP}, I_{RD}, I_{FC}, I_{FD}) and each of the nine indicator products (1–9): e.g. $I_{PP1\text{-}9}$ ... specific impacts of a product in the primary production step of the chain for each of the nine indicator products

To determine the amount of eaten food (m_F), the amount of food waste was subtracted from the domestic utilization of food (m_U) in the consumption step ($m_{FW/FC}$) (Eq. (1)).

$$m_F = m_U - m_{FW/FC} \quad (1)$$

Table 2
Total amount of food production and food domestic utilization in EU 2011 (excl. alcohol) per food category and share of indicator product.

Food category	Indicator product category	Production in EU 2011 (excl. alcohol)		Food domestic utilization in EU 2011 (excl. alcohol)	
		Billion kilos/year	Share of indicator product within food category	Billion kilos/year	Share of indicator product within food category
Cereals		293,091	100.0%	63,367	100.0%
	Bread from cereals	293,091	100.0%	63,367	100.0%
Roots & tubers		62,383	99.9%	36,549	99.8%
	Potatoes and products	62,298	99.9%	36,473	99.8%
Oil crops, pulses, sugar crops & nuts	–	214,431	0.0%	35,286	0.0%
Fruits & vegetables		130,353	21.5%	109,511	21.2%
	Tomatoes and products	16,261	12.5%	13,918	12.7%
	Apples and products	11,717	9.0%	9302	8.5%
Meat & animal fat		58,675	74.5%	49,761	79.3%
	Pig meat	23,374	39.8%	20,489	41.2%
	Poultry Meat	12,285	20.9%	11,003	22.1%
	Bovine Meat	8059	13.7%	7947	16.0%
Fish & seafood		6735	28.2%	11,597	31.6%
	White fish (Demersal)	1897	28.2%	3663	31.6%
Dairy & eggs		162,410	95.8%	127,927	95.2%
	Milk – Excluding Butter	155,527	95.8%	121,839	95.2%
Other (miscellaneous)	–	143	0.0%	10,797	0.0%
Total products		928,221		444,795	
Total indicator products		584,509		288,001	
Share of indicator products		63.0%		64.7%	

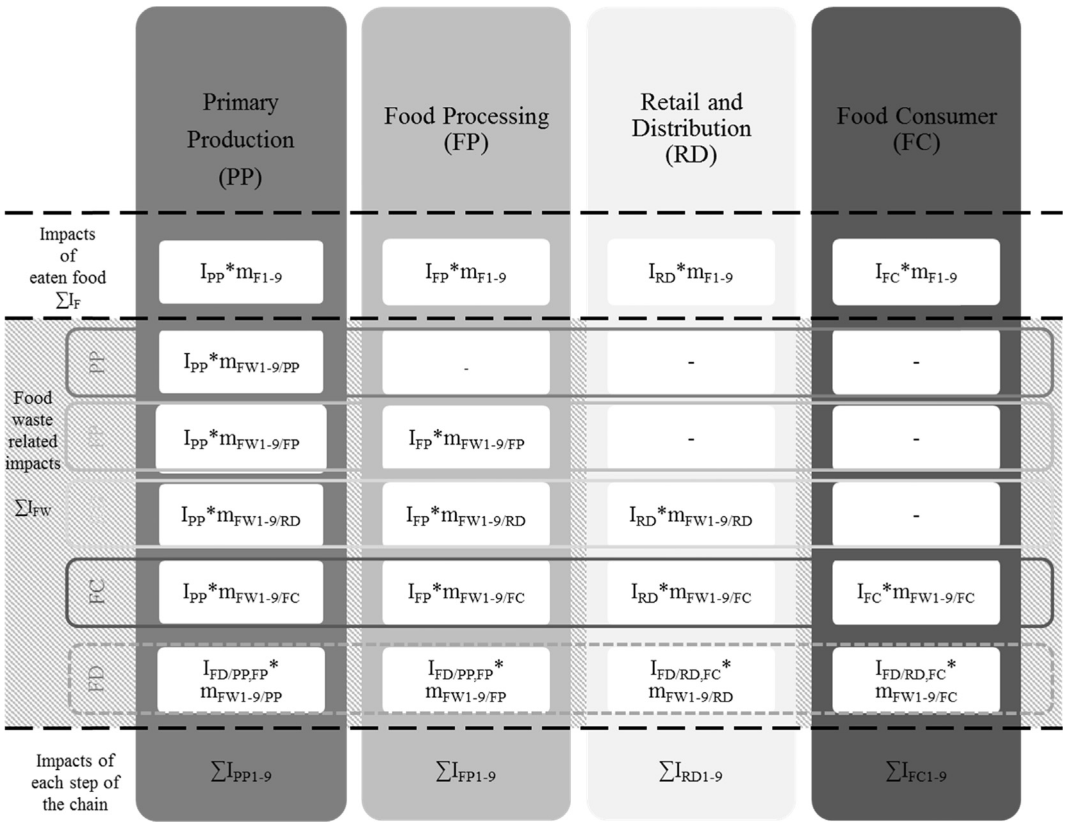

Fig. 2. Parameters for the calculation of environmental impacts of food eaten by the consumer and wasted along the supply chain.

The overall impacts of food and food waste in this study consist of food eaten by the consumer and food wasted along the supply chain. The parameters are furthermore illustrated in Fig. 2 to show the allocation of impacts to certain steps of the chain.

Impacts of eaten food derive from impacts in each step of the chain multiplied by the mass eaten from the consumer (see Eq. (2)).

$$\text{Impacts of eaten food}: \quad I_F = (I_{PP} + I_{FP} + I_{RD} + I_{FC})m_F \tag{2}$$

Food waste related impacts include those created throughout the supply chain plus the impacts from food waste management (food disposal). The further along the supply chain a product is wasted, the higher the impacts caused. Such waste also drives further food production. To determine the sum of food waste related impacts (I_{FW}), impacts at each step of the chain need to be calculated. Eq. (3) shows the example of impacts generated in the primary production step which can be attributed to food waste. The total food waste related impacts which originate from the primary production step ($I_{FW/PP}$) comprise the sum of specific impacts of the product in the production step ($I_{PP1\text{-}9}$) multiplied by the sum of specific mass which is wasted along the supply chain of this product (as all food which is wasted also needs to be produced) ($m_{FW1\text{-}9/PP} + m_{FW1\text{-}9/FP} + m_{FW1\text{-}9/RD} + m_{FW1\text{-}9/FC}$) plus the specific impacts of food disposal ($I_{FD/PP,FP}$) multiplied by the specific mass of food waste at production ($m_{FW1\text{-}9/PP}$).

$$\Sigma I_{FW/PP} = I_{PP1-9} * m_{FW1-9/PP} + I_{PP1-9} * m_{FW1-9/FP} \ldots\ldots + I_{FD/PP,FP} * m_{FW1-9/PP}$$

or

$$\Sigma I_{FW/PP} = I_{PP1-9} * (m_{FW1-9/PP+} m_{FW1-9/FP+} m_{FW1-9/RD+} m_{FW1-9/FC}) + I_{FD/PP,FP} * m_{FW1-9/PP} \tag{3}$$

The sum of food waste related impacts generated in each step of the chain (Eq. (4)), show us the food waste related impacts along the whole supply chain.

$$\Sigma I_{FW} = \Sigma I_{FW/PP} + \Sigma I_{FW/FP} + \Sigma I_{FW/RD} + \Sigma I_{FW/FC} \tag{4}$$

2.6. Environmental impact categories

The selection of environmental impact categories involves an identification of categories which are not only of relevance to the LCA study (ISO, 2006a, 2006b) but also for which sufficient evidence can be found relating to the chosen food products. Defra (2011) classified impact categories from existing LCAs on food products across thirteen known environmental impacts. Four of the environmental impact categories, Global Warming Potential (GWP), Eutrophication Potential (EP), Acidification Potential (AP) and Reported Energy (RE), showed sufficient reported information with GWP being the most widely reported impact category. This is also examined in LCAs on European consumption (e.g. Tukker et al., 2006) where studies covering GWP and AP agree that food is one of the major contributors. When considering eutrophication, food dominates all other consumption products and services (Tukker et al., 2006). Other food consumption and food waste related studies also used GWP, AP and EP as impact categories (see Gruber et al., 2015). This work presents results on GWP measured in carbon dioxide equivalents (CO_2-eq.), AP measured in sulphur dioxide equivalents (SO_2-eq.) and EP measured in phosphate equivalents (PO_4-eq.).

2.7. Extrapolation of results

The extrapolation of the results for each food product to the total food eaten by the consumer and food waste, make the incorporation of an extrapolation factor necessary. The basis for extrapolation in this study is domestic food utilization in EU 2011. In a first step an up-scaling from indicator product to food category is carried out (see share of indicator products within each food category in Table 2). In a second step extrapolation from six covered food categories to eight categories is conducted to incorporate the categories oil crops and pulses as well as other miscellaneous with the factor 1.12 (398,712 billion kilos/year are covered by six chosen food categories, to up-scale to a total of 444,795 billion kilos/year a factor of 1.12 is necessary). It is assumed that the indicator products have the same impacts as the products not considered.

Regarding the time representativeness, the bottom-up approach faces different years of data origin. The mass flows for food waste are based on data from 2012 (see Stenmarck et al., 2016). However, the basis for extrapolation is domestic food utilization in EU 2011. According to ILCD (European Commission, 2010) the represented year of a data set shall be determined by looking at the different ages of the main contributing data. The results in this assessment can hence be referred to the year 2011, as domestic utilization is the main contributing data.

3. Life cycle inventory

3.1. Mass flows (m)

3.1.1. Food waste on product level ($m_{FW1\text{-}9}$)

The quantities of food waste produced at each step of the food supply chain are based on the FUSIONS food waste levels by Stenmarck et al. (2016). According to the dataset the total food waste produced in the EU is 88 Mt, distributed along the supply chain as follows: 9 Mt in primary production, 17 Mt in processing, 5 Mt in wholesale and retail, and 57 Mt in households and food service.

The bottom-up approach demands data on product level. As the food waste data set which was used as a basis show data only on an aggregated level, a further step to disaggregate first to food category level (e.g. fruits and vegetables) and then to indicator product category level (e.g. apples and products) was necessary. To generate mass flows at the food category level, the data and approach by Gustavsson et al. (2013) was applied. Gustavsson et al. (2013) used Food Balance Sheets by FAOStat (FAOStat, 2011) and literature-derived data on food waste amounts, and calculated total food waste amounts for each food supply chain step and for each food category. The relative shares of food waste within each food group (i.e. how much each product group contributes to the total food waste amount) of Gustavsson et al. (2013) was used and applied to disaggregate the food waste dataset of Stenmarck et al. (2016). Mass flows on indicator product level were generated by using the information of domestic food utilization (relative shares of each product within its food category). So, is the food category 'fruits and vegetables' represented by 'apples and products' and 'tomatoes and products' in the same proportion for both supply and waste side. Table 3 shows the quantity of food waste per indicator product in each step of the supply chain. It can be seen that consumption step has highest shares in each of the indicator products. This is due to the high amount of waste generated at consumer level, which reveals the food waste data set from Stenmarck et al. (2016). Cereals are based on Gustavsson et al. (2013) the highest contributor to food waste, followed by dairy products, such as milk, and potatoes, which is also shown in Table 3.

3.1.2. Waste management options on product level

In addition to food waste data at the product level, the identification of treatment and disposal routes of food waste in the EU was necessary. As data on e.g. vegetal waste in Eurostat cover also other waste apart from food waste (e.g. green waste) this statistical dataset could not be considered without further adaptation. Additionally, and more importantly, food waste is often disposed together with other waste fractions such as residual waste, so that the waste treatment options are varied and often depend on the national waste management strategy as well as on the extent of waste separation at source. These circumstances make further adaptions of Eurostat data necessary to receive a complete picture of the EU's status of food waste collection and treatment. Furthermore, a major distinction should be made between animal-containing food waste and vegetable-only waste as those waste streams are handled differently in food waste management.

The main data source used to detect waste management operations in Europe was provided by Eurostat, and was combined with selected national statistics and research studies and articles from scientific journals to help to fill data gaps. Eurostat's categories for waste treatment operations were merged with food waste treatment options within the FUSIONS definitional framework from Östergren et al. (2014).

For the consideration of the extent of separate collection of biodegradable waste in EU-28, information from national reports of the ECN (2015) and the report of Saveyn and Eder (2014) were employed. The information revealed that 14 European countries have separate collection systems installed, but the extent varies considerably. While in the Netherlands 90% of the households are covered by separate collection of biodegradable waste, there is a separate collection installed only for larger housing units or in some pilot regions in some Eastern European countries. As a baseline it was assumed that 65% of the EU member states have a separate collection installed with a participation rate of 40% of households. The magnitude of separate collection activities

Table 3
Food waste on the level of indicator products in 1000 tonnes calculated from aggregated food waste levels of Stenmarck et al. (2016) and shares of each food category on the basis of Gustavsson et al. (2013).

Indicator product category	Production	Processing	Retailing	Consumption	Total
Apples and products	363	103	175	1489	2131
Tomatoes and products	504	116	245	1891	2755
Potatoes and products	2331	3593	496	4172	10,592
Cereals - Excluding Beer	832	6577	527	17,026	24,962
Milk - Excluding Butter	861	1352	259	9527	11,999
Bovine Meat	43	367	137	904	1452
Pigmeat	126	948	336	2337	3746
Poultry meat	66	527	187	1300	2080
White fish (Demersal)	17	98	67	179	361
Indicator product TOTAL waste	5143	13,681	2429	38,825	60,078
Total food waste by Stenmarck et al. (2016)	9100	16,900	4600	57,000	87,600
Share indicator products from total food waste	57%	81%	53%	68%	69%

remains a crucial factor to detect the current practice of food waste management in Europe. The impact on the overall environmental effects will be discussed later. Furthermore, Eurostat does not differentiate between anaerobic digestion and composting. Both options are covered under the term 'Recovery other than energy recovery'. From information found in ECN (2015) it was assumed that 80% of separate collected food waste is sent to composting facilities and 20% to anaerobic digestion.

Table 4 illustrates the allocation of waste management operations for different waste streams: food waste mixed with other household and similar waste, vegetal food waste and animal containing food waste. In the mixed food waste stream, a certain amount of separate collection at source was incorporated, which increases the contribution of composting and anaerobic digestion compared to Eurostat data. Yet, in Europe, landfilling is still the most prominent option for disposing of food waste mixed with household waste. Another picture emerges for animal containing and vegetal food waste which originates mostly from the production and processing industry and is not mixed to household or household similar wastes. Those food waste streams are mostly treated via composting facilities, and to a lesser extent by anaerobic digestion. One category of waste disposal highlighted in Eurostat concerns backfilling. This involves disposing of waste in excavated areas such as underground mines or gravel pits (Eurostat, 2013), but is not generally relevant for the disposal of food waste.

Table 4
Shares of waste treatment operations for different food waste streams in Europe.

Waste treatment operations	Animal and mixed food waste	Vegetal food waste	Food waste mixed to other household and similar waste (incl. also sep. collection)
Composting	67.3%	74.5%	27.9%*
Anaerobic digestion	16.9%	18.7%	7.0%*
Co-generation	5.7%	3.4%	17.0%
Incineration	2.1%	0.4%	11.2%
Landfill	7.3%	2.8%	36.9%
Plough-in/not harvested, Sewer, Discard	0.6%	0.1%	0.0%
Backfilling	0.1%	0.0%	0.1%

* Share is increased by the part of food waste which is covered by separate collection in Europe.

3.2. Environmental impacts

Data inventories used for the calculation of environmental impacts are shown in Tables 5–7 for each indicator product and each stage of the supply chain. Data of production, processing, retail and distribution are based on existing LCA studies, whereas data of consumption and waste management were added manually to this assessment to generate the same system boundaries. The data sources used for each stage and the given assumptions are explained in the sub-chapters.

3.2.1. Data inventory of primary production, processing, retail and distribution (I_{PP}, I_{FP}, I_{RD})

Data from existing LCA studies were collected via a comprehensive literature review using both scientific literature and general sources to search for evidence and data, guidance and metrics to define the environmental impacts of the selected indicator products. Sources included Google, Web of Science™, Scirus, scientific journals and reviews (both peer and non-peer reviewed), book chapters and company information/websites. In total 134 sources

Table 5
Median values of global warming potential in kg CO_2-eq./kg product flow.

Indicator product	I_{PP}	I_{FP}	I_{RD}	I_{FC}	$I_{FD/PP,FP}$	$I_{FD/RD,FC}$
Apple	1.0E−01	2.5E−02	1.8E−01	9.1E−02	6.1E−03	1.7E−01
Tomato, field	2.5E−01	1.0E−01	3.3E−01	1.1E−01	6.1E−03	1.7E−01
Tomato, greenhouse	2.1E+00	1.0E−01	1.5E−01	1.1E−01	6.1E−03	1.7E−01
Tomato, total	7.2E−01	1.0E−01	2.9E−01	1.1E−01	6.1E−03	1.7E−01
Potato, fresh	1.5E−01		1.2E−01	6.9E−01	6.1E−03	1.7E−01
Potato, frozen	2.5E−01	5.3E−01	2.2E−01	9.2E−01	6.1E−03	1.7E−01
Potato, chilled	4.4E−01	7.9E−01	5.8E−01	2.9E−01	6.1E−03	1.7E−01
Potato, total	1.9E−01	2.0E−01	1.6E−01	7.6E−01	6.1E−03	1.7E−01
Bread	5.5E−01	1.4E−01	1.3E−01	9.1E−02	6.1E−03	1.7E−01
Milk	1.1E+00	8.3E−02	1.2E−01	1.1E−01	2.7E−02	1.7E−01
Beef	2.8E+01	4.9E−01	4.4E−01	1.3E+00	2.7E−02	1.7E−01
Pork	5.9E+00	2.0E−01	6.3E−01	1.1E+00	2.7E−02	1.7E−01
Chicken	3.0E+00	2.1E−01	4.6E−01	1.1E+00	2.7E−02	1.7E−01
Fish	2.6E+00	1.6E−01	9.3E−01	4.0E−01	2.7E−02	1.7E−01

Table 6
Median values of Acidification Potential in g SO2-eq./kg product flow.

Indicator product	I_{PP}	I_{FP}	I_{RD}	I_{FC}	$I_{FD/PP,FP}$	$I_{FD/RD,FC}$
Apple	4.2E−01		9.4E−01	5.1E−02	−3.5E−04	−9.2E−05
Tomato, field	1.3E+00		1.6E+00	5.1E−02	−3.5E−04	−9.2E−05
Tomato, greenhouse	2.8E+00		1.2E+00	5.1E−02	−3.5E−04	−9.2E−05
Tomato, total	1.7E+00		1.5E+00	5.1E−02	−3.5E−04	−9.2E−05
Potato, fresh	1.5E+00	0.0E+00	6.8E−01	1.9E−01	−3.5E−04	−9.2E−05
Bread	4.2E+00	5.0E−01	5.0E−01	0.0E+00	−3.5E−04	−9.2E−05
Milk	1.3E+01	1.2E−01	5.5E−01	5.1E−02	−3.1E−04	−9.2E−05
Beef	3.3E+02	5.4E−01	1.6E+00	3.0E+00	−3.1E−04	−9.2E−05
Pork	8.2E+01	5.4E−01	2.0E+00	2.3E+00	−3.1E−04	−9.2E−05
Chicken	6.9E+01	9.3E−01	1.1E+00	2.8E+00	−3.1E−04	−9.2E−05
Fish	29.64	0	2.3E+00	2.5E−01	−3.1E−04	−9.2E−05

Table 7
Median values of Eutrophication Potential in g PO4-eq./kg product flow.

Indicator product	I_{PP}	I_{FP}	I_{RD}	I_{FC}	$I_{FD/PP,FP}$	$I_{FD/RD,FC}$
Apple	2.4E−01		1.3E−01	1.5E−02	5.7E−06	4.7E−04
Tomato, field	4.7E−01		2.6E−01	1.5E−02	5.7E−06	4.7E−04
Tomato, greenhouse	6.3E−01		2.6E−01	1.5E−02	5.7E−06	4.7E−04
Tomato, total	5.1E−01		2.6E−01	1.5E−02	5.7E−06	4.7E−04
Potato, fresh	7.7E−01	0.0E+00	8.5E−02	9.0E−03	5.7E−06	4.7E−04
Bread	2.1E+00	9.0E−02	9.6E−02	0.0E+00	5.7E−06	4.7E−04
Milk	6.3E+00	2.0E−01	1.5E−01	1.5E−02	7.0E−05	4.7E−04
Beef	1.4E+02	4.0E−01	2.4E−01	3.2E−02	7.0E−05	4.7E−04
Pork	4.2E+01	4.0E−01	3.5E−01	2.4E−02	7.0E−05	4.7E−04
Chicken	2.7E+01	2.7E−01	1.2E−01	3.0E−02	7.0E−05	4.7E−04
Fish	2.7E+00	2.7E+00	3.9E−01	7.6E−04	7.0E−05	4.7E−04

were collected, which provided information on GWP (108 sources), on EP (72 sources) and AP (71 sources) for one or more indicator products. The full references of the various sources are shown in the published report of Scherhaufer et al. (2015).

The outcome of the literature study was a database of environmental impact (GWP/AP/EP) values for each stage of the food supply chain. The arithmetic averages, median values, first quartiles and third quartiles as well as the standard deviations were calculated to detect the data correlations within each indicator product. If major discrepancies between data within indicator products occurred a weighting was applied to the values. In the case of potatoes, major differences concerning GWP were found between fresh and semi-prepared products. As a consequence, a weighting was carried out according to the UK retail value sales of the potato market in 2011 to adjust the data: 63% fresh, 33% frozen, 2% chilled and 2% canned/dehydrated (canned potatoes were distributed between frozen, chilled and fresh due to the lack of data). Impact data for different potato products for EP and AP was unfortunately lacking and was represented by fresh potato. In case of tomatoes major differences appear between greenhouse and field tomatoes. It was assumed that 75% were field-grown and 25% were produced by greenhouse cultivation across Europe. Impacts from transport processes and from packaging materials are attributed to the step 'retail and distribution'.

If literature data showed only aggregated results for some or all of the product life cycle stages, an estimation for a segregation by life cycle stage is considered in proportion to the other references for the same product providing data for each life cycle stage. Where possible the impacts reported were converted to common units of impact characterisation and scaled to the functional unit.

Existing LCA data mostly consist of life cycle phases from cradle to farm gate. The consumption and end of life phase were integrated in this study to cover all fields within the system boundaries.

For further calculations within the bottom up approach the median values were considered, which are shown in Table 5, Table 6 and Table 7.

3.2.2. Data inventory of food consumption (I_{FC})

Most of the consumer based activities were not covered in the literature (e.g. cooking, consumer travel) and had to be added with simplified assumptions.

The cooking behaviour of consumers is different for some of the indicator products. The following considerations were taken in accordance with publications of Defra. In case of potatoes where no data was available, it was assumed that fresh potatoes are boiled for 30 min on 1500 W hob (0.75kWh equivalent to 0.345 kg CO_2eq/kg), chipped/prepared potato products are oven cooked for 25 min in 2000 W oven (0.83 kWh equivalent to 0.382 kg CO_2-eq/kg) and mashed potato is heated in 1500 W microwave oven for 8 min, (0.2 kWh equivalent to 0.09 kg CO_2eq/kg). For pork and pork products, the following assumptions were applied to the various data points: pork meat was oven cooked for 90 min at 180 °C in 2000 W oven (therefore 3kWh equivalent to 1.38 kg CO_2eq/kg); minced pork/sausages is fried/cooked for 30 min on 1500 W hob, therefore (0.75 kWh equivalent to 0.345 kg CO_2eq/kg); bacon is fried for 10 min on 1500 W hob (therefore 0.25 kWh equivalent to 0.115 kg CO_2eq/kg); tenderloin is oven cooked for 60 min at 180 °C in 2000 W oven (therefore 2 kWh equivalent to 0.92 kg CO_2-eq/kg). Beef was assumed to be oven cooked for 80 min at 180 °C in a 2000 W oven (therefore 2.66 kWh equivalent to 1.23 kg CO_2eq/kg) and Chicken was assumed to be cooked for 75 min at 180 °C in a 2000 W oven (therefore 2.5 kWh equivalent to 1.15 kg CO_2-eq/kg). Literature data was used for the greenhouse gas emissions from food retailing for potatoes, chicken, pork and beef (Defra, 2008).

For consumer travel, a transport of, on average, 10 kg of food by passenger car over a distance of 5 km was assumed for each of the indicator products. Storage in a refrigerator was added to the consumption emissions for tomatoes which was considered to be similar to the literature value found for milk.

The influence of the given assumptions at consumer level are discussed later.

3.2.3. Data inventory of food disposal ($I_{FD/PP,FP}$ and $I_{FD/RD,FC}$)

If food disposal was covered in existing LCAs, it was found to be mostly about manure management. Other waste management operations for food waste which is generated in production and processing and also in the consumption step are not covered in the selected literature. For this reason, impacts related to manage and treat food when it is wasted, so called food waste management, were included in the assessment and, hence, double-counting does not apply.

The division between food waste which is separately collected and food waste which is mixed with other waste fractions is crucial, when it comes to the calculation of environmental impacts from waste management operations as the routes are different (see Table 4). Whether food waste is disposed of separately or mixed with other waste streams often depends on the food waste generator and the waste management practices utilised in the individual countries. Based on common waste management practices in Europe, specific food waste flows can be allocated to different generators:

- Production and processing phases generate animal and mixed food waste (in case of milk, beef, pork, chicken, fish) and vegetal food waste (in case of apple, tomato, potato, bread) respectively. Impacts from the food disposal in these phases are allocated to the waste management options for animal and mixed food waste and vegetal food waste ($I_{FD/PP,FP}$).
- Wholesale, logistics, retail and marketing, in-home and out-of-home consumption phase generate food waste which is mixed

to other household and similar waste. A certain magnitude of separate collection at source is incorporated. Separate collected biodegradable waste ends up either at composting or anaerobic digestion plants. Impacts from the food disposal in these phases are allocated to the waste management options for food waste mixed to other household and similar waste ($I_{FD/RD,FC}$).

Impacts may vary between the indicator products as the products contain different percentages of dry matter and volatile solids as well as the specific production factor of methane (see Eriksson et al., 2015). In this study the impacts of overall biodegradable waste were considered instead. Impact data were taken from GaBi 6.0 database of ThinkStep (formerly PE International GmbH) and in case of composting and anaerobic digestion of Pertl and Obersteiner (2014). Carbon within food is of biogenic origin and accounted as carbon neutral. Therefore, only methane, nitrogen dioxide and fossil carbon are taken into account in case of the Global Warming Potential.

For the composting process, composting of biodegradable waste in a closed system was considered with environmental benefits from substitution of fertilizer (nitrogen, phosphorous and potassium) and peat. Anaerobic digestion of biodegradable waste was chosen with composting of the digestate in a closed system incl. spreading of digestate (substitution of fertilizer and peat, and of electricity from the grid mix and heat from fossil fuels). For co-generation, data from the GaBi database (waste incineration of biodegradable waste fraction in municipal solid waste) with substitution of electricity from the grid mix (EU-27 electricity mix 2002) and heat (EU-27 District Heating Mix 2002) was considered. For incineration where energy output is not used, those substitutions were not assessed. For the landfilling process, data from the GaBi database (landfill of biodegradable waste) was considered and for plough-in/not harvested or discarded, data of a rotting process of biodegradable waste in open windrow composting from Pertl and Obersteiner (2014) was taken as an assimilation process to cover the emissions. Credits for material recovery in case of plough-in/not harvested food (e.g. for avoided use of fertilizers because of returned nutrients to the field) and carbon sequestration were not considered in Pertl and Obersteiner (2014) and are therefore also not covered in this study. As the importance of backfilling for food waste disposal is very low (see Table 4), impacts from backfilling were neglected.

4. Results

4.1. Environmental impacts of eaten food and food waste

Impacts on global warming, eutrophication and acidification occur because of emissions along the entire supply chain of food. Environmental impacts are shown in Figs. 3–5 for emissions allocated to the amount of food which is eaten and those which can be allocated to food which is wasted within the EU. A total amount of 88 Mt of food waste in the EU (Stenmarck et al., 2016) results in 186 Mt CO_2-eq. relevant for global warming potential, 1.66 Mt SO_2-eq. relevant for acidification potential and 0.74 Mt PO_4-equivalents relevant for eutrophication potential. The share of food waste related impacts to impacts of the whole supply chain is 15.7% for GWP, 15.1% for AP and 15.2% for EP. As mass flows are the same in each of the considered impact categories, the share of food waste impacts to overall impacts differs due to impacts only from food disposal. Also since the impacts of food disposal have a very low contribution to overall impacts, this does not affect greatly the overall scale.

Most of the food waste related impacts are generated by emissions in the production step of the chain, which is shown in the bar charts of Figs. 3–5. In case of GWP 73.4% of the impacts can be attributed to the production step. Those impacts gain a greater share when it comes to AP (94%) and EP (96%). Regarding GWP (Fig. 3) the processing step has a contribution of 5.6%, retail and distribution of 7.4%, food consumption of 7.9% and finally disposal of 5.7%. Regarding AP (Fig. 4), processing and consumption have a contribution of 1.2% and 1% respectively, whereas retail and distribution is higher with 3.4%. AP shows a negative value for the impacts from food disposal. That is due to credits given for composting and anaerobic digestion and the use of its outputs as fertilizer as well as credits for co-generation. Those credits also affect the other environmental impact categories, but not in such a high share as when considering AP.

Considering EP (Fig. 5) the share of processing, retail and distribution is 1.5% and 1.3% and of consumption only 0.1%. The share of food disposal is less than 0.01%.

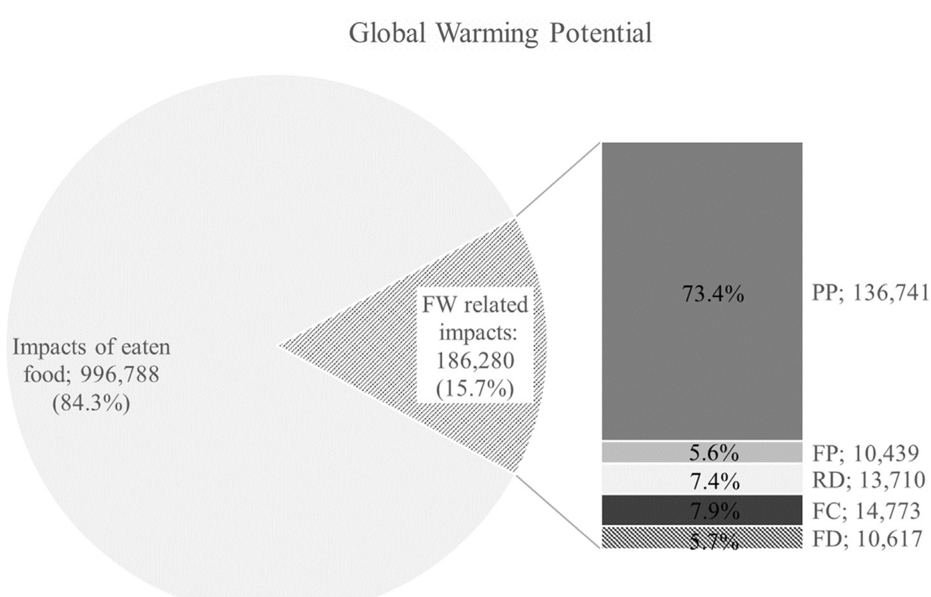

Fig. 3. GWP of eaten food and food waste (left) and food waste related impacts generated at different steps of the chain (right) in 1000 tonnes CO_2-eq in the EU.

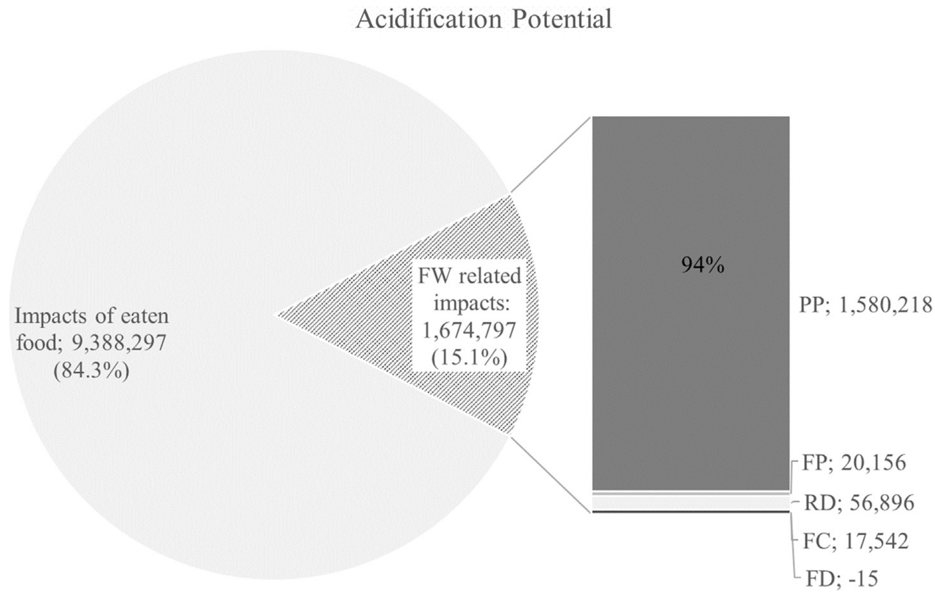

Fig. 4. AP of eaten food and food waste (left) and food waste related impacts generated at different steps of the chain (right) in tonnes SO2-eq in the EU.

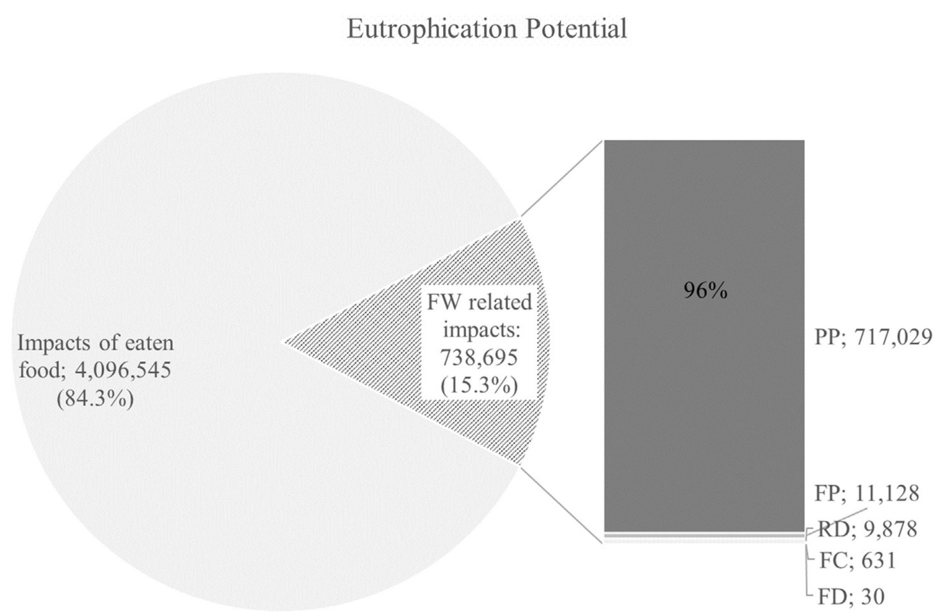

Fig. 5. EP of eaten food and food waste (left) and food waste related impacts generated at different steps of the chain (right) in tonnes PO4-eq in the EU.

4.2. Food waste related impacts on product level

The environmental impacts of total food eaten including food waste show that most of the impacts derive from the production step. This applies for most of the products except for potato and apple with regard to GWP (see Fig. 6). In case of the potato the consumption step gains more importance due to the long boiling time and the respective energy use in this activity. In case of the apple the transport included at retail and distribution is proportionately higher due to the low impacts along the rest of the apple supply chain. Even though bread is the most relevant stream in food waste by mass, beef shows the highest food waste related impacts followed by pork in all three categories.

AP also shows the high influence of the production step in most of the indicator products (see Fig. 7). However, retail and distribution plays also an important role in case of apple, tomato and potato. This can be allocated to transport processes within retail and distribution which gain more importance when the impacts of production are comparably lower. Impacts of retail and distribution do not feature highly here.

Considering EP, the primary production step of the food supply chain gains even more importance due to leakages of nutrients in agriculture (Fig. 8). An exception is the indicator product fish, where the processing step also plays a significant role. This can be explained by different system boundaries being reported for the production & processing steps and it may depend on whether the processing is carried out on the vessel or in a factory. It may also be related to discards on the vessel.

The contribution of each indicator product to total food waste related impacts is shown in Fig. 9 for each environmental category.

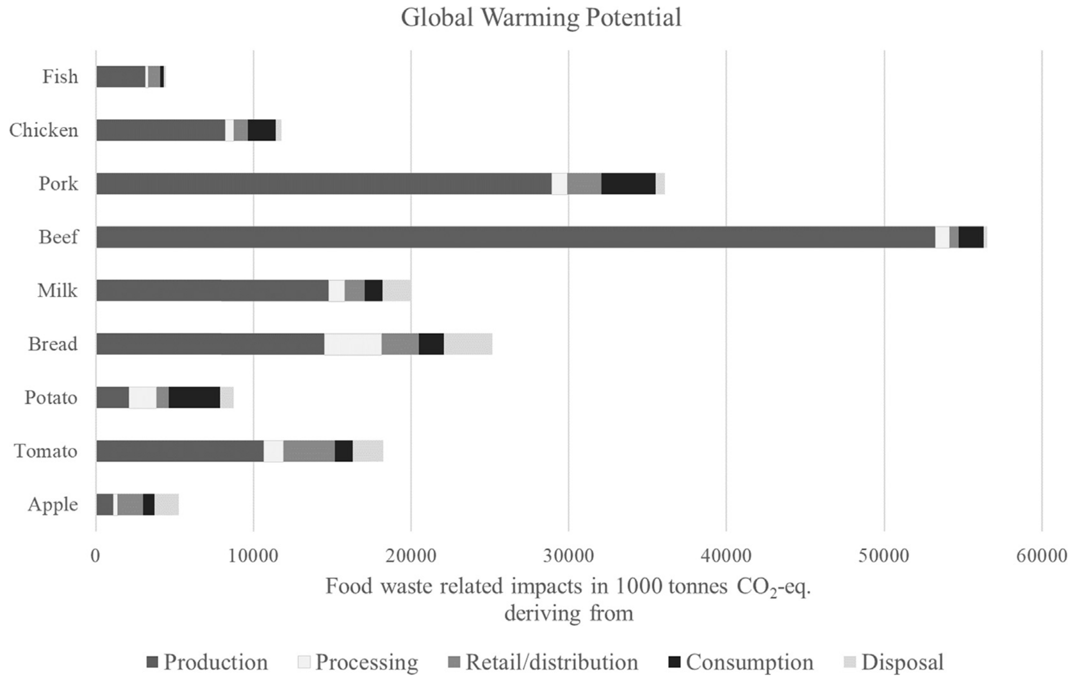

Fig. 6. GWP of food waste related impacts per indicator product and steps where the impacts derive from in 1000 tonnes CO_2-eq in the EU.

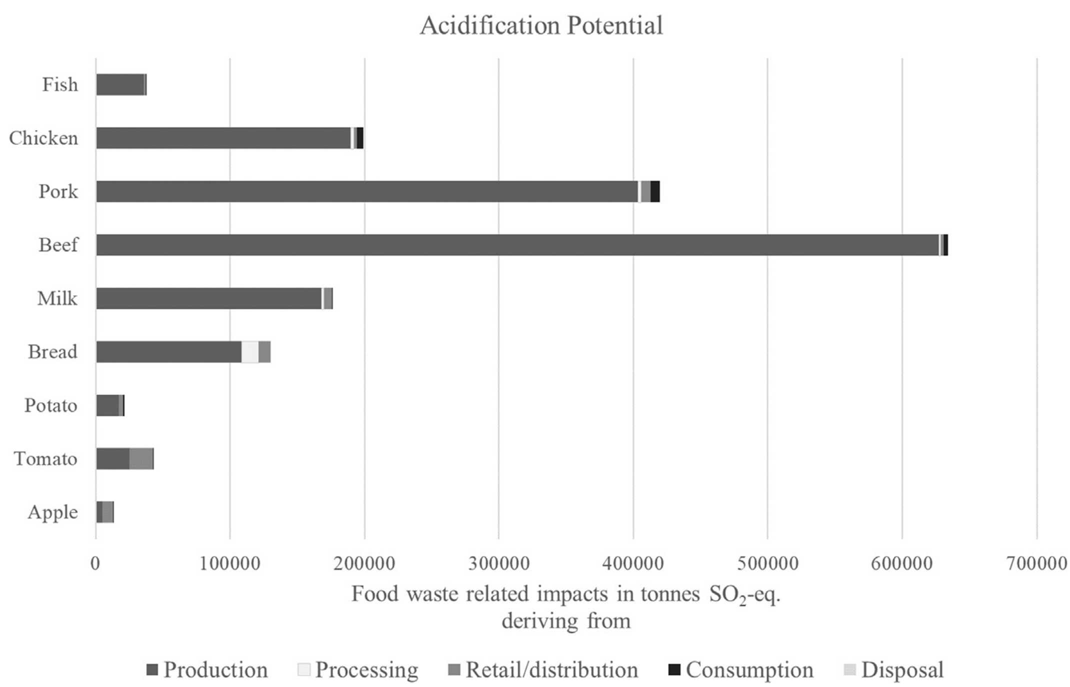

Fig. 7. AP of food waste related impacts per indicator product and steps where the impacts derive from in tonnes SO_2-eq in the EU.

Meat and dairy products have a share of 69% considering GWP, 88% considering AP and 89% considering EP to total food waste related impacts. Vegetables and fruits have lower shares. Noticeably higher values within meatless food are observed for bread and tomato: bread, because of its high amount by mass found in food waste and tomato, due to the high energy requirements for greenhouse tomato production which comes into effect when considering GWP.

5. Discussion

5.1. Relevance of food waste related impacts

According to current estimates from Stenmarck et al. (2016) Europe produces 88 Mt of food waste each year. This amount is set in relation to the amount eaten by the consumer in Europe. If the domestic food utilization in the EU is used minus food

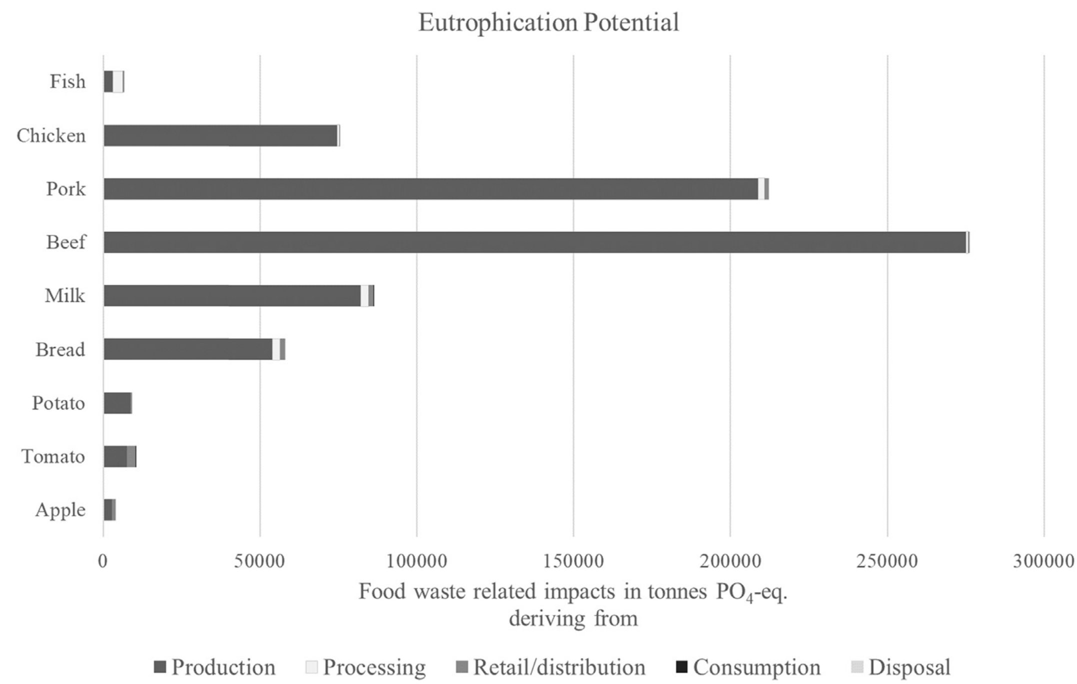

Fig. 8. EP of food waste related impacts per indicator product and steps where the impacts derive from in tonnes PO_4-eq in the EU.

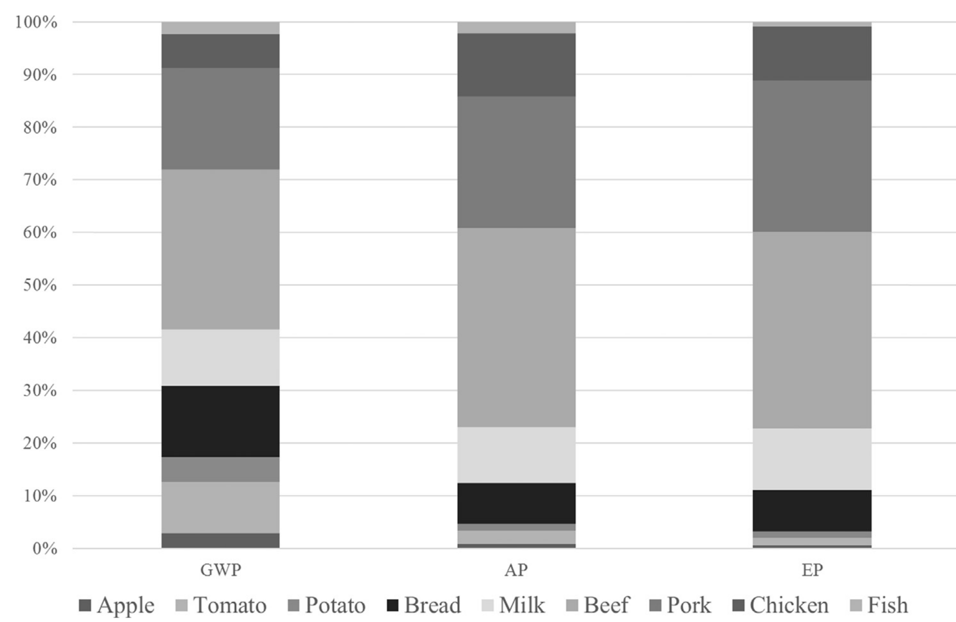

Fig. 9. Share of impacts of each indicator product to total food waste related impacts per environmental category.

waste at the consumption step, it results in a total amount of food eaten by the consumer of 388 Mt. In order that a consumer can eat 1 kg of food, 1.226 kg need to be produced, this corresponds to a total loss of 18.4% along the entire food supply chain. When it comes to environmental impacts, the share differs slightly. Food waste related impacts are 15 to 16% of the total impacts from the entire supply chain depending on the environmental category used. This means that the share of environmental impacts from food waste is lower than the share by mass. This can be explained by the reason, that the most dominant food found in waste are cereals (see Table 3). Bread, which represents in this study the food category cereals, has lower impacts on the environment compared to other products (see data inventory in Tables 5–7). Products which have higher environmental impacts such as meat products (see data inventory in Tables 5–7), are often consumed in lower amounts. However, meat and dairy products still cause a lot of impacts in all considered environmental categories even if they are lower by mass compared to other products (see Fig. 9). This is also acknowledged in the studies of Monier et al. (2010) and FAO (2013). A reduction in meat and dairy products in food waste would significantly reduce the environmental impacts.

5.2. Comparison of results

Impacts on Global Warming Potential resulted in an estimated 186 Mt of CO_2-eq. emitted in Europe because of food waste. Those impacts derive from supply chain emissions of food which is wasted instead of being eaten plus all emissions coming from food disposal activities such as composting or landfilling. This result is within the same range as the estimations based on Monier et al. (2010) with 170 Mt CO_2-eq., but far lower than the estimations from FAO (2013) with around 500 Mt CO_2-eq. This is due to different mass flows used in the studies. When regarding the impacts per tonne of food waste, one can see that the results of this study, with around 2.13 t CO_2-eq./t food waste, is between Monier (1.9 t CO_2-eq./t food waste) and FAO (approx. 3.6 t CO_2-eq./t food waste). This difference can be due to several reasons (methodology, different system boundaries, different data basis). A comparison in detail to the results of this study is therefore limited.

However, within the project FUSIONS a top down approach was furthermore carried out in line with the bottom-up approach which uses the same dataset and system boundaries. This comparison revealed insights to the plausibility of results with regard to greenhouse gas emissions and further points of research. The comparison of results via two different approaches is common for consumption related LCAs (such as Tukker et al., 2006 in comparison with Huysman et al., 2016 as well as Quested and Johnson, 2009) and was therefore found to be suitable also for food waste related impacts. The top-down approach carried out in the EU FUSIONS project to detect the environmental impacts of food waste resulted in 2.93 t CO_2-eq./t food waste (Scherhaufer et al., 2015), which is higher than the results shown by the bottom-up approach. To pinpoint the differences in results of those two approaches it is necessary to look at the shares of food waste related impacts per life cycle step. The shares of the bottom-up approach are 73.4% of food waste related impacts coming from primary production, 5.6% from processing, 7.4% from retail and distribution, 7.9% from consumption and 5.7% from disposal. The top-down approach revealed 65.4% from production, 10.7% from processing, 5.9% from retail and distribution, 10.7% from consumption and 7.3% from disposal. Impacts from the production step are therefore lower in the top-down approach when relative shares are considered. The processing step shows nearly double the impacts as detected via the bottom-up approach. Also the consumption step shows a higher relative share. These indications make it necessary to look at these steps of the chain in detail. Explanations for the difference and recommendations for future studies are summarised in the following sections.

5.3. Sensitivity check

A range of uncertainties are inherent with the approach used in this study: using a numerous amount of emission factors from existing LCA studies, mass flow data with moderately high uncertainties and certain scenario assumptions. The assessment is mainly based on existing LCA studies in this study. No LCA according to ISO (2006a, 2006b) was performed. Therefore, an uncertainty calculation was not conducted. The appropriateness of the collection and selection of inventory data (referred to as 'data uncertainties') as well as the scope and scenario choices (referred to as 'scenario uncertainties') are instead critically discussed in this section.

5.3.1. Data uncertainties

Environmental impact results shown in this study are based on mass flows of food domestic utilization and food waste within the EU. The quantification of food waste in the EU is still lacking in many steps of the chain, countries and level of detail (food category or even food product level). The data of Stenmarck et al. (2016) is based on existing studies on food waste quantification of EU member states which are in line of the chosen definition of 'food waste' and which show plausible and reliable results. Data were only obtained for up to a quarter of member states and the process of scaling the information from these member states to the whole EU-28 is responsible for a relatively high data uncertainty. The necessity of finding a clear and robust way of food waste quantification is unquestionable. Tools to quantify food waste have already been developed (e.g. FAO's food loss and waste protocol or FUSIONS' food waste manual) to improve data quality and completeness. A more accurate data basis on food waste quantities will result in a more accurate determination of environmental impacts.

Another point is the lack of food waste data at product level, which is, however, demanded when using a bottom-up approach. Data on the composition of food waste is only available on food category level (e.g. fruits) and not on product level (e.g. apple). To generate the product level, consumption data was used. The composition of food which is eaten may differ from the composition of food which is wasted. For example, within the category fruits, the most consumed product may be apples, but the most wasted product may be a pineapple, which have completely different environmental impacts.

The uncertainty and lack of data applies also for the type of food e.g. fresh, frozen or preserved food, which have different environmental impacts. The indicator products used in this study are rather fresh food than processed food (except in the case of potato), which may underestimate the impacts from the processing step. This is acknowledged when comparing the results with the top-down approach, which shows a higher share of emissions from the processing step. Foster et al. (2006) highlights the importance of incorporating heavily processed foods (frozen or preserved) in future LCA studies. However, information regarding the frequency of such products in food waste, is still a data gap. The influence of processed food on the environmental impacts is recommended for further investigation as soon as more data on product level is available.

Furthermore, assumptions had to be taken at the consumer and disposal step of the food chain to show impacts of an EU average situation. The consumption step and with it the consumer behaviour with regard to transport, storing and cooking has significance in this study when it comes to GWP (share of 8% to overall food waste related impacts) and to potato and meat products (more than 10% of its impacts), because of the high energy demand for cooking. Gruber et al. (2015) already mentioned that consumer behaviour has a great influence in LCAs on food. The fuel consumption for consumer travel has the highest effects in her study. Gruber et al. (2015) considered consumer travel of 10 km by car, whereas this study assessed 5 km by car. Consumer travel is definitely an issue, which needs to receive further attention, especially when it comes to an estimation of an average consumer behaviour of EU-28, which was not the focus of this study. Furthermore, the storage and cooking behaviour are of relevance according to Gruber et al. (2015), which is also acknowledged in this study. These issues should be addressed in future studies.

Furthermore, mass flows going to different waste management options require a further improvement of existing data. It is still not entirely clear which pathway is used in individual EU member states for food waste coming from different steps of the food supply chain. The extent of separate collection of food waste clearly influences the waste management option. Separately collected food waste are more feasible for recycling options such as anaerobic digestion or composting than food waste which is mixed with other waste types, such as residual waste from households. The impact factor of vegetal food which is treated in composting or anaerobic digestion plants is, regarding GWP, 27 times less than for food waste which is mixed to other waste fractions and sent mostly to landfill (see Table 5). If food waste valorisation including the separate collection of food waste could be implemented more widely across the EU-28, the impact reduction potential in this step could be significant. However, the contribution of food waste treatment and disposal operations to the total food waste related impacts is very low (6% GWP). Even though most of the food waste occurs at consumption step (65%, Table 3) and most of the food waste from the consumption step still ends up at landfills in Europe (40%, Table 4), the impacts are not relevant compared to the impacts created by supply chain emissions. Additionally, this study used a high share of landfilling as a baseline, which will phase out in near future (as soon as the landfill directive is implemented in all of the MS which bans the landfilling of untreated biodegradable waste) and therefore the relevance of food waste treatment compared to impacts from other steps of the supply chain will even be lower in future.

5.3.2. Scenario uncertainties

Scenario uncertainties which need to be addressed are the choice of system boundaries, the way of solving multi-functionality, bundling impact results out of a range of secondary literature and finally the choice of environmental impact categories.

It needs to be highlighted that in this study only certain aspects of the food supply chain are assessed and are therefore within system boundaries. Food which is removed from the supply chain and fed to animals is not part of this study. Furthermore, impacts which are coming from the treatment of human excretions are also beyond this study. The impact of eaten food would likely be higher if this were considered. The relation between food waste related impacts to food which is eaten by EU citizens need to be interpreted with caution as it does not show the impacts of the whole food supply chain. Additionally, regarding system boundaries, emissions of packaging, if contained in used datasets, was included in the process of retail and distribution, but wasn't separately displayed in the results. According to Silvenius et al. (2014) the significance of packaging is relatively low for climate change, eutrophication and acidification compared to the production chain of certain food products. However, it is without question that packaging can support food waste prevention. The type of packaging can influence the product life time and consequently reduce food being wasted. From an environmental point of view, the reduced impacts due to food waste prevention can offset higher impacts due to more sophisticated packaging systems (Denkstatt, 2014; Williams and Wikström, 2011).

The issue of solving multi-functionality is topic of a range of studies. Laurent et al. (2014) reviewed 222 published LCA studies on solid waste management systems and revealed the use of 'all imaginable' types of solving multi-functionality in past practice. The differentiation between context situations which demand attributional or consequential modelling described in the ILCD guidelines (European Commission, 2010) are not always clear. This study rather follows an attributional approach with existing interactions with other systems which can make a system expansion and substitution approach necessary (see p. 43 of European Commission, 2010). No additional future consequences are addressed in this modelling. As no large-scale consequences are addressed in this approach, average data for substituted products is legitimately used. However, it is also a point of discussion, especially if the threshold of large-scale consequences is reached by food waste reduction measures. A revised version of ISO 14044 hopefully will bring more specific guidance on solving multi-functionality, which has already been recommended in studies such as Pelletier et al. (2015).

Another important aspect, though, is the issue that several different characterization methods (CML, EDIP, etc.) can be used for the calculation of environmental indicators such as GWP, EP or AP. The characterization method used in diverse LCA studies is not always stated. Most of literature used in this study mention CML (Guinée, 2002) as characterization method, but some also use ReCiPe (Goedkoop et al., 2009). Furthermore, methods can be adapted to updated values from e.g. IPCC to calculate GWP which result in different standards used for certain impact categories. Those aspects may have an influence on the results. However, the influence can hardly be quantified. Instead the authors decided to take the median value instead of an average (mean) value, which may narrow down the most typical impact factor for a specific product. The strength of this approach is that a range of studies is considered, the weakness is that the selection of the studies may not be suitable to serve as a typical value. For this reason, a comparison with a recently carried out study by Clune et al. (2016) was conducted. Clune et al. (2016) completed a meta-analysis of LCA studies for different fresh food categories with in total 369 published studies. The GWP values in Clune et al. (2016) were converted to the system boundary of the Regional Distribution Centre. Consequently, the values can only be compared from production to retail and distribution in this study. The median values of Clune et al. (2016) and this study in those three steps of the chain are very similar when comparing animal containing product categories. Major differences appear only in the food category 'cereals' with 0.5 kg CO_2-eq./kg product, whereby this study uses the indicator product 'bread' with 0.83 kg CO_2-eq./kg. The choice of two indicator products representing the food category vegetables can also be underlined with the results by Clune et al. (2016). Clune et al. (2016) determined a median GWP value for field grown vegetables of 0.37 kg CO_2-eq./kg produce, whereas this study used a higher GWP value of 0.68 kg CO_2-eq./kg product for tomato in combination with a lower GWP value of 0.27 kg CO_2-eq./kg for potato. Another difference occurred in the category 'fruits' (0.42 kg), where this study only used 'apple' (0.31) as an indicator for this group. The differences are legitimate as the current study focus on the most consumed product within a food category and not on an average value representing the whole food category. As a conclusion, the authors find it legitimate to use the median value out of a range of secondary literature for the purpose of this study. The comparison with Clune et al. (2016) shows that the chosen values or the literature behind those values are not chosen in an arbitrary manner. Nevertheless, GWP values found for the categories 'vegetables' and 'fruits' can have a wider range, which might gain more importance, when more details on food waste product level is available (e.g. if pineapple is found to be the most waste food within the food category 'fruits' instead of apples which is the most consumed food) as mentioned in Section 5.3.1.

For EP and AP the situation may be different. For example, the EP depends on an effect factor to algae which can differ from study to study (see also Seppälä et al., 2004). However, the influence on the results is not easy to quantify here. This is definitely a point of further research.

Although there are uncertainties coming along with this approach, it still serves as an approximate estimation of the impacts associated with food waste. Global warming, eutrophication and acidification are relevant impacts in relation to the emissions from the food supply chain. However, the impacts shown in this study do not include land use change (LUC), which is another important aspect when it comes to food. Bellarby et al. (2013) states that a great contribution to LUC comes from dairy products as dairy cows are fed using soy-beans. Audsley et al. (2009) determined that UK's food consumption leads to an increase of food's footprint by 50% if LUC is considered. The contribution of land use change to EU's food waste is definitely a point of further research. However, the link between food and land use change is not entirely clear. Effects of land use change need to be addressed on global level with equal responsibilities as land use changes (direct and indirect) are due to an overall global pressure on land. Models on indirect LUC can be found for example in Schmidt et al. (2015) and Tonini et al. (2016).

6. Summary and conclusions

This study presents indicative estimates of the potential scale of the food waste related impacts based on available food waste data on European level. The impacts were calculated via nine indicator products (apple, tomato, potato, bread, milk, beef, pork, chicken, fish) from the bottom-up to the total amount of food eaten and food waste in the European context. Food waste in this study is understood to be any edible or inedible part of food removed from the supply chain and sent to food waste treatment and disposal facilities. Food which is donated or fed to animals is not covered here, as they are considered as food or by-product but not as food waste. Impacts from food waste throughout the food supply chain and food waste management are estimated to be 186 Mt CO_2-eq, 1.7 Mt SO_2-eq. and 0.7 Mt PO_4-eq. The share of food waste related impacts to the overall impacts from food eaten by the consumer is 15.7% regarding GWP, 15.1% regarding AP and 15.2% regarding EP. As mass flows are the same in each of the considered impact categories, the share of food waste impacts to overall impacts is only differing due to impacts from food disposal. On the level of indicator products, bread with 34% has the highest contribution of food waste related impacts to total impacts which is due to the high amount of cereals found in food waste. Most of the food waste related impacts derive from the primary production step of the chain. Regarding GWP, food waste related impacts can be attributed with 73% to the production step, 6% to food processing, 7% to retail and distribution, 8% to food consumption and 6% to food disposal. The contribution of impacts out of production to overall food waste related impacts is even higher when AP and EP are considered (94% and 96% respectively). The biggest contribution to overall food waste impacts come from meat products (Fig. 7), even if food waste is dominated by cereals by mass (Table 2).

As uncertainty calculation or sensitivity analysis were not performed it was necessary to discuss uncertainties on a qualitative basis. The interpretation of results revealed that more attention should be paid to impacts from the processing industry and from consumer behaviour, which would call for the inclusion of more semi-prepared, prepared and frozen food products in the bottom-up approach and a more detailed assessment of the consumer behaviour of EU-28 member states. The bottom-up approach serves as a suitable way of identifying hotspots to mitigate global warming and other impact categories. Additionally, the approach allows for the inclusion of additional products, such as semi-prepared food, in the assessment to get a more concrete picture of current impacts related to food waste. However, this implies the availability of food waste data in the necessary detail, which is currently lacking on a European basis.

Despite lacking data, certain key messages can be derived from the results of this study. The hotspots of food waste lie in the emissions caused during the supply chain and in the type of food which is wasted. Impacts from food waste treatment and disposal are not the driving factor of food waste related impacts. Most of the impacts derive from the primary production step. Priority should therefore be given to food waste prevention. By preventing food from being wasted, emissions in any subsequent stages of the food supply chain can be avoided. Priority should also be given to preventing meat and dairy products from food waste, as they cause most of the impacts. Food waste prevention at household level has the greatest effect to mitigate global warming. This is, firstly, because a high share of potentially avoidable food waste occurs at household level in terms of mass and, secondly, because the impacts at the consumer stage include all accumulated impacts from earlier stages of the supply chain. Ambitious but necessary prevention measures will be to raise consumer awareness with the goal of a real consumer change. However, not all of the food waste can be prevented. A specific share of food waste is not avoidable (e.g. the inedible parts of food). For food waste which is not avoidable, the most sustainable treatment option should be chosen. Although the contribution of food waste treatment and disposal is not relevant compared to overall emissions, there are major differences in the type of treatment option. Landfilling has the highest environmental impacts compared to other treatment options such as anaerobic digestion or composting and therefore should only be used as the option of last resort. If food waste cannot be prevented, then it should at least be separately collected in order to enable a suitable food waste treatment or food waste valorisation.

Facing the challenge to mitigate global warming, which is addressed by the Paris Agreement ratified by the European Commission, food waste prevention can play a significant role. The quantification of food waste by mass and also by its environmental impacts on European level is an important data baseline to set targets and to identify hotspots which need to be prioritised by measures to fight food waste.

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